Second Research of Astra 6 Extra High

Mechanistic estimation of random neural networks | Second-round research response

My assessment: the aggregate family deserves one inexpensive feasibility test, but I do not currently see a complete path to the leaders' cost. The most informative new observable is the rank of its **metric-side query**, which imposes a constraint that M3's spectrum of U does not test.

All costs below are float32 units, with rounded leading terms. Proposed experiments use the supplied checkpoint; I have not run them on your network.

Q1. Fit an aggregate cubic, with an early rejection test

Derived - mathematical object. Use the existing admissible old-content subspace, without reducing its rank:

$$\widehat{T} = \sum_{a=1}^{k} \mathrm{sym}_3\left[(QS_aQ^\top) \otimes (Qu_a)\right], \qquad S_a = S_a^\top \in \mathbb{R}^{r\times r}.$$

Allow indefinite S_a. Components have no birth identity. This is a linear-times-quadratic decomposition, associated with polynomial *strength*, rather than ordinary CP rank. Bik, Draisma and Eggermont.

Its geometry is restrictive. Writing $V=\mathrm{span}(u_1,\ldots,u_k)$,

$$\widehat{T}(z,z,z) = 3\sum_a (z^\top S_a z)(u_a^\top z) \quad \Longrightarrow \quad \widehat{T}|_{V^\perp\times V^\perp\times V^\perp} = 0.$$

Conversely, a cubic vanishing on $V^\perp$ admits such a decomposition: every monomial contains at least one coordinate from V. Thus "a few metric-vector components" requires a large subspace on which the cubic vanishes approximately. Low rank of individual legs does not establish this property.

Derived - a caveat concerning the closed CP experiment. A component with $\mathrm{rank}(S_a)\le r$ has symmetric CP rank at most 2r+1, using

$$a^2V = \frac{(a+v)^3-(a-v)^3-2v^3}{6}.$$

Consequently, k=4, r=384 implies CP rank at most 3076<4n. This proposal survives the CP measurement only if that measurement concerned a particular column-capping procedure, not the globally best unrestricted CP approximation. If the latter was established, small-k aggregation is already excluded.

Derived - transport and readout. Between insertions, compute

$$BQ = Q^+R, \qquad S_a^+ = RS_aR^\top, \qquad u_a^+ = Ru_a.$$

This transport is exact before basis truncation or recompression. For query rows $Z\in\mathbb{R}^{m\times r}$, define

$$H_a(Z) = \mathrm{diag}(Zu_a)\,ZS_a, \qquad K_a(Z) = d_a u_a^\top, \qquad (d_a)_i = z_i^\top S_a z_i.$$

Then the symmetric tensor's quadratic-vector query is

$$J(Z) = \sum_a [2H_a(Z) + K_a(Z)].$$

At Z=Q, D21=J(Q)$Q^\top$, and D3=$3\Sigma_a d_a \circ (Qu_a)$. No dense leg products occur.

Q1, continued. Cost and fitting procedure

Derived - running cost. Including explicit thin QR, transport, readout and one final expansion, the leading cost between insertions is

$$c_{\text{layer}} \approx 2a + 2a^2 - \tfrac{2}{3}a^3 + k(a^2 + 2a^3), \qquad a = r/n.$$

At r=384, this is approximately **1.00+0.25k units/layer**, including a small elementwise allowance.

But incoming sources generally lie outside the transported basis. Basis maintenance cannot be free. For example, a one-pass Gram range finder incorporating BQ, A_{new}, P_{new}, followed by projection and explicit QR, gives approximately **2.92+0.25k units per insertion layer**, before fitting. Extra power iterations increase this.

With 250−107=143 units retained elsewhere and approximately eleven old-content layers:

- Even the optimistic k=4 calculation consumes about **165 units before fitting**.
- The specified basis-refresh implementation reaches approximately **186 units before fitting**.
- Storing full dense G_a instead costs approximately 2k units/layer; even k=1 leaves essentially no fitting allowance at 165 units.

These are implementation estimates, not impossibility bounds for every absorption algorithm.

Measure-this - fitting procedure and first falsifier.

1. Test the necessary rank condition first. To fit your four source-formula queries separately, use two component families corresponding to the AAP and MPP terms. These are term types, not source identities. For either family,

$$K(Q) = DU^{\top}, \qquad \operatorname{rank} K(Q) \le k_{\text{family}}.$$

Therefore its optimal rank-k_{family} tail must satisfy

$$\frac{\sum_{j > k_{\text{family}}} \sigma_j(K)^2}{\|K\|_F^2} \le 4 \times 10^{-4}.$$

Form the two exact metric-side siblings in n×384 coordinates and compute their singular spectra. Allow **under 8 units** from stored factors/legs. If their required ranks sum to more than four, reject the proposed small-k, four-query fit immediately. Crucially, H can have high rank even for one component; **M3's rank-230 U therefore does not directly imply k=230**.

2. If that screen passes, use alternating least squares. Minimize the sum of the four normalized squared query errors, requiring each final relative error to be at most 0.02.

Train on the current 1024 query rows plus 256 rows from one specified later linear map. With u_a fixed, solve jointly for symmetric S_a using matrix-free conjugate gradients. With S_a fixed, solve jointly for u_a. Normalize u_a, absorbing scale into S_a. Initialize u_a from the metric-side right singular vectors; also try a random initialization.

A matrix-block normal-operator application costs approximately

$$2qk(r/n)^2 = 0.352k$$

units, where q=1280/1024. Five alternating rounds, twenty matrix-CG iterations per round, k=2,4, and two starts require about **422 units** for those applications. Budget **500 units including vector solves, targets and validation**: about eighteen minutes at sustained 1 GFLOP/s.

Q1, continued. Validation and interpretation

3. Validate without refitting on independent query rows and rows from the actual next transport. Failure there rejects the fitted state, even if current-layer error passes. A failed nonconvex optimization does not certify that every rank-k fit fails.

Important qualification: separately matching four implementation-specific terms is stronger than matching their symmetric sum. A different representation can redistribute contributions between those terms. Thus the rank screen conclusively rejects the *requested four-query fit*, not every representation reproducing total D21. If allowing that redistribution, run the same fitting procedure on J, retaining transported holdouts.

Conjecture - expected k. M1-M3 do not identify it. My working expectation under the strict four-query objective is $O(10^2)$ components per family if the metric-side spectra resemble the measured mixed-side spectrum, rather than two to four total. The proposed spectrum measurement replaces that conjecture cheaply.

Derived - accumulated error. Linear transport introduces no new model error, but transports existing error through $B^{\otimes 3}$. Insertion errors add after different transports; they need not cancel. A 2% fit at one gate therefore does not guarantee 2% subsequently. Successful gate fits must eventually be evaluated cumulatively.

Q2. Other formulations: distinct state or relocated tensor?

Derived - characteristic or cumulant-generating functional. Under the linear stage,

$$\psi^+(t) = \psi(B^\top t).$$

Retaining the cubic part exactly retains the same symmetric tensor. Cubic ridge features reproduce CP transport; linear-times-quadratic features reproduce Q1. An unrestricted cubic feature basis in r=384 dimensions contains

$$\binom{386}{3} = 9{,}511{,}040$$

coefficients. Ordinary dense core rotation plus query evaluation costs approximately **115 units/layer**, before births.

A nonlinear functional representation could differ, but no specified feature family here has both cheap composition under $B^\top$ and a closed ReLU update. I would not propose another random-feature experiment after M3.

Derived - predictable tensor covariance plus a small realized odd state. Let $t=\mathrm{vec}_{\mathrm{sym}}(T)$. Its formal updates are

$$t^+ = \mathcal{B}t + s, \qquad \Sigma^+ = \mathcal{B}\Sigma\mathcal{B}^\top + \text{birth/cross terms},$$

where the operator represents $B^{\otimes 3}$. Even statistics can describe uncertainty about t, but cannot select its realized odd value. A signed linear decoder with d coefficients needs its omitted **query-covariance** eigenvalues to contain at most 0.04% of query energy. One matrix U's singular spectrum is not that covariance spectrum.

Cheap isotropic statistics do not supply those coefficients. Their moving basis again needs a transport representation: CP, a full core, or something like Q1. A full unrestricted covariance would contain roughly 9×10^{13} entries at r=384. I see no distinct affordable update here.

Q2, continued. Responses and contraction order

Derived - response functions. Exponentially tilt the activation distribution by $e^{z^{T}h}$. Then

$$\frac{\partial C_{ij}(z)}{\partial z_k}\Big|_{z=0} = T_{ijk}, \qquad C^{+}(z) = BC(B^{\top}z)B^{\top}.$$

This is a useful interpretation, but the derivative obeys exactly third-order transport. Restricting its directional dependence to k vectors gives matrix-vector components again, costing approximately **2k units/layer** in full coordinates, or Q1's projected costs. Calling them Jacobian statistics does not remove the insertion problem.

Derived - birth-first contractions and diagram resummation. Contracting birth indices first produces the already-closed cubic core: approximately **54 units per r=384 core construction** in the ordinary contraction order. A TAP or diagrammatic method would be genuinely new only if it supplied an explicit small family of quenched vertices closed under transport and source insertion. Covariance-only or trace-only vertices are already excluded. I cannot provide such a family with an $O(n^2)$ update.

These are rejections, not recommendations to repeat the closed experiments.

Q3. One hypothesis about the leaders

Conjecture: they use a **different coupled closure**, whose errors in third-order transport are compensated by its treatment of higher-order terms, rather than accurately reproducing V29's intermediate D21.

This is not the rejected learned output correction: the compensation would occur inside the evolving moment equations. Nevertheless, I cannot name a cost-complete implementation, so this is an explanation to test, not a proposed solution.

Measure-this: obtain one leader's layerwise third-order corrections. If they differ substantially from V29, replace only those corrections with V29's values while retaining the leader's remaining closure equations. **Worsening final MSE after that replacement would support coupled error compensation.** Agreement within 2% instead would favor a genuinely efficient representation.

With executable leader code and existing ground truth, one V29 run plus baseline/intervention leader runs costs at most approximately **580 units**, around twenty-one minutes at 1 GFLOP/s.

Nothing currently presented constitutes a plausible complete solution at 133-165 units. The strongest obstruction is the combination of pervasive old-content sensitivity, failure of existing compressions, and the tiny allowance left for both aggregate transport and source absorption. The new metric-side rank test can determine, for under eight units, whether the surviving strict aggregate fit merits further work.